Formalizing Interpretable Kernel-Level Security: A Multimodal Analysis of Neural Rule Distillation and eBPF Enforcement in Cloud-Native Infrastructures

The rapid shift toward microservice-based architectures has fundamentally altered the threat landscape of modern computing environments. As containers share the host operating system's kernel, the primary boundary for security enforcement has moved from the network perimeter to the system call interface. While traditional intrusion detection systems (IDS) relied on static signatures, the emergence of deep learning—particularly Transformer architectures—has enabled the detection of sophisticated, multi-stage attacks by modeling the long-range dependencies inherent in system call sequences.[1] However, the deployment of high-capacity neural models in production environments is frequently hindered by their computational overhead and opaque decision-making processes. To address these challenges, the current research landscape has focused on the intersection of Knowledge Distillation and Explainable Artificial Intelligence (XAI). By training a Transformer "Teacher" to identify complex behavioral anomalies and subsequently distilling this logic into white-box "Student" rules, security practitioners can generate interpretable policies that are suitable for real-time enforcement at the kernel level using technology such as the extended Berkeley Packet Filter (eBPF).[4] This report provides a comprehensive analysis of the state-of-the-art literature across six critical domains, formalizing the transition from black-box neural detection to verifiable, low-latency kernel security.

Area 1: Transformer and Deep Learning Architectures for System Call Intrusion Detection

The application of Transformer models to system call analysis represents a departure from the limitations of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, which often struggle with the vanishing gradient problem when processing the extremely long traces typical of modern container workloads.[1] The self-attention mechanism, the core of the Transformer, allows for parallelized processing and the capturing of non-linear relationships across thousands of events.[1]

Transformer-Based Research in System Call

Sequences

Paper Title	Authors	Year	Venue	Summary and Relevance
Next-Generati on IDS: Leveraging Transformer and Hybrid Deep Architectures for Robust Cyber Threat Detection	Anonymous	2024	ResearchGate	This work introduces a deep learning-based architecture using Transformer networks to recognize long-range dependencies in system call sequences. It is highly relevant as it establishes the Transformer's ability to overcome sequencing noise and generalize to new attack patterns compared to traditional HIDS.[2]
MulAD: A Framework for Detecting Complex Anomalies in Distributed Systems	Anonymous	2025	Science of Computer Programming	MulAD integrates sequential, semantic, quantitative, temporal, and parametric log patterns using

				a Transformer/Bi-LSTM hybrid. It specifically addresses log interleaving by rearranging sequences based on source parameters, which is critical for multi-threaded container environments.[10]
TransIDS: An Interpretable Hybrid Neural Network Architecture for Anomaly Detection	J. Wu et al.	2025	IEEE MASS	The paper proposes a multi-modal approach combining packet-based Transformers with log-based language models to identify anomalies. It demonstrates how attention mechanisms can be used to weight the importance of specific log messages, aiding the interpretability of the classification.[11]

BERT-Based Intrusion Detection System for RF Jamming Attacks in Vehicular Network	N. R. Wickramasurendra	2024	Master's Thesis (Brock University)	This study evaluates pre-trained Transformer architectures (BERT, RoBERTa, DistilBERT) for multi-class attack classification. It provides a blueprint for using BERT-style embeddings to capture subtle patterns in cybersecurity datasets while optimizing for resource efficiency.[12]
An Enhanced Transformer-Based Intrusion Detection Model	Anonymous	2023	Applied Sciences (MDPI)	The researchers propose a two-stage learning strategy where the model first performs binary prediction of an intrusion and then refines the classification into specific categories. This hierarchical

				approach is relevant for filtering high-volume syscall traffic before complex analysis.[7]
Real-time Network Intrusion Detection via Decision Transformers	Anonymous	2023	Hugging Face Papers	This work casts real-time intrusion detection as a causal sequence modeling problem using Decision Transformers. It highlights a trade-off between detection accuracy and timeliness, which is a core consideration for kernel-level deployment.[13]
DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning	M. Du et al.	2017 (Cited in 2024)	ACM CCS / ArXiv	While pre-dating 2019, DeepLog is the foundational baseline for modeling logs as natural language using LSTMs. It introduces the concept of

				separating log entries from concurrent tasks into different sequences, a precursor to modern interleaving solutions.[14]

The handling of multi-thread interleaving is a recurring theme in the transition from simple host-based IDS to containerized environments. As nodes perform tasks in parallel, chronological ordering alone becomes insufficient.[10] Methods like parameter-based reconstruction ensure that the Transformer "sees" a coherent execution path for each process. Furthermore, the use of semantic embeddings (e.g., mapping syscall names to a 768-dimensional latent space) allows the model to understand the functional similarity between different kernel operations, such as read() and recvfrom(), which may be used interchangeably by an attacker.[15]

Encoding and Tokenization Mechanisms

The effectiveness of the Transformer "Teacher" is heavily dependent on the encoding of raw kernel events. Tokenization often involves representing each syscall as a unique token, but recent works emphasize the importance of positional embeddings and disentangled attention.[1] DeBERTa, for instance, isolates the representation of content and relative position, which prevents the model from being confused by the fragmented payloads common in modern exploits.[16]

Feature Type	Implementation	Relevance to Master's Thesis	Source
Lexical Features	Script length, character distribution, entropy.	Basic filtering before neural processing.	16
Semantic Embeddings	WangChanBERTa/BERT-style latent manifolds.	Capturing deep intent of syscall sequences.	15

Structural Features	Standardized global syntactic properties.	Identifying deviations in expected program flow.	[16]
Positional Embeddings	Sinusoidal or learned position markers.	Essential for maintaining syscall order.	[1]

Area 2: Explainability and Rule Extraction from Neural Networks

The inherent complexity of Transformer models—often characterized by millions of parameters and multi-head attention layers—makes them "black-box" systems unsuitable for forensic accountability.[18] The research area of Explainable AI (XAI) addresses this by developing methods to extract human-understandable logic from trained models, often through surrogate modeling or post-hoc attribution.[20]

Rule Extraction and Surrogate Modeling Research

Paper Title	Authors	Year	Venue	Summary and Relevance
Interpreting Unsupervised Anomaly Detection in Security via Rule Extraction	Anonymous	2024	ResearchGate	This paper proposes a post-hoc method to globally explain unsupervised models via the "Interior Clustering Tree," which decomposes complex data into human-unders tandable rule sets. It directly

				supports building surrogate models for online deployment in high-stake security applications.[22]
Learnable Knowledge Distillation (LKD): Automating the Design of Distillation Loss	Anonymous	2025	PMC	LKD employs a bi-level optimization strategy to autonomously learn adaptive distillation losses aligned with the student model's validation performance. This is crucial for ensuring the fidelity of rules extracted from a complex Transformer teacher into a simpler student.[4]
Global Counterfactual Rules for Actionable Network Intrusion Detection	Anonymous	2025	Applied Sciences (MDPI)	This work generates actionable global rules from counterfactual explanations, summarizing

				important features and value bounds for attack detection. These rules are designed to filter out incoming queries, providing a direct path to eBPF implementation.[23]
Intrusion Detection in IoT Using CNN: An Explainable AI Approach	Ebrahimi et al.	2025	Cybersecurity (Springer)	The study evaluates SHAP, LIME, and RuleFit for producing local and global interpretations of IDS decisions. It emphasizes the need for separate interpretations for each attack type, which is a gap the thesis can fill by applying this to syscalls.[24]
SCAR: Stealthy Distillation-Conditional Backdoor Attack	Anonymous	2025	NeurIPS / ArXiv	This research explores the security vulnerabilities of the

				Knowledge Distillation process itself, showing how backdoors can be reactivated during distillation. It serves as a warning for verifying the integrity of the "Teacher" model before rule extraction.[25]
Model Extraction Attacks on Large Language Models	Anonymous	2025	ArXiv	This paper analyzes how surrogate models can be trained to replicate a target model's functionality through query-response patterns. It provides technical details on the three phases of distillation: query generation, response collection, and surrogate training.[27]

Knowledge Distillation (KD) acts as the primary mechanism for transferring the Transformer's learned decision boundaries into interpretable formats such as Decision Trees (DT) or Rule Lists.[4] The "Teacher" model provides soft labels (logits) which contain more information than

binary hard labels, allowing the "Student" to learn the relative relationships between different types of malicious and benign behavior.[25] This process is mathematically represented by the minimization of the Kullback-Leibler (KL) divergence between the teacher and student output distributions:

$$\mathcal{L}_{KD} = \sum_i p(y_i|x;\theta_T) \log \frac{p(y_i|x;\theta_T)}{p(y_i|x;\theta_S)}$$

where θ_T and θ_S represent the parameters of the teacher and student models, respectively.[4]

Post-Hoc Methods and Fidelity Metrics

Post-hoc explanation methods like SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) are widely cited for identifying feature importance.[28] However, for the purpose of a thesis focusing on kernel deployment, these methods are often viewed as diagnostic tools rather than deployment candidates due to their high computational cost.[29] Fidelity metrics—measuring how closely the extracted rules match the teacher's original predictions—are the standard for evaluating these surrogate models.[22]

Method	Interpetability Level	Computational Cost	Suitable for eBPF?	Source
Decision Trees	Global & Local	Low	Yes (C-translatable)	30
RuleFit	Global	Moderate	Yes (Linear ensemble)	21
SHAP	Local	High	No (Post-hoc only)	29
Anchors	Local	Moderate	Limited (Condition-heavy)	21
Counterfactuals	Local/Actionable	High	Yes (as Global Rules)	21

Area 3: eBPF-Based Runtime Security and Policy Enforcement

The transition from rules to enforcement requires a technology capable of executing logic at kernel speeds without introducing prohibitive overhead. eBPF has revolutionized this domain by providing a verifiable, high-performance sandbox within the Linux kernel.[5]

Comparative Analysis of eBPF Security Tools

The current literature provides significant data on the performance and architectural trade-offs of the most prominent eBPF security tools (ESSTs).[33]

Feature	Falco	Tetragon	Tracee	KubeArmor	Source
Architecture	Rule-based Alerting	Policy Enforcement	Forensic Logging	Hardening	[33]
CPU (Attack)	433.5 mcore	6.9 mcore	93.7 mcore	N/A	[33]
Memory (Attack)	392.08 MB	641.50 MB	570.47 MB	N/A	[33]
MTTD (Escape)	High	1.178 s (Fastest)	Moderate	N/A	[33]
Action Type	Alert Only	Block/Enforce	Log/Audit	Block/Restrict	[34]

Automated Policy Generation and In-Kernel Machine Learning

Recent research has explored the feasibility of deploying machine learning models directly within eBPF programs to bypass the overhead of context switching between kernel and userspace.[5]

Paper Title	Authors	Year	Venue	Summary and Relevance
A Flow-Based IDS Using Machine Learning in eBPF	M. Bachl et al.	2021	ArXiv	This seminal work proves that Decision Trees and simple Neural Networks can be implemented entirely in eBPF. It reports a 20% performance increase compared to userspace implementations, directly supporting the thesis's goal of kernel-level deployment.[31]
SmartX Intelligent Sec: Intelligent Security Framework Leveraging eBPF/XDP and Bi-LSTM	Anonymous	2024	ArXiv	The framework uses eBPF/XDP for packet capturing and filtering while utilizing a Bi-LSTM for threat detection. It demonstrates a dual approach: mitigating threats in real-time while maintaining high detection

				accuracy.[6]
Comparative Analysis of eBPF-Based Runtime Security Monitoring Tools in Kubernetes	Syairozi & Arizal	2025	RITECH	Provides a rigorous KPI comparison of Falco, Tetragon, and Tracee across OWASP Kubernetes Top 10 threats. It concludes that Tetragon is superior for CPU efficiency and detection time in container escape scenarios.[33]
Real-time Cryptojacking Container Attack Detection Framework	Anonymous	2025	Electronics (MDPI)	This paper describes a framework using Tetragon to capture syscall traces and n-gram sequences for machine learning training. It achieves 99.75% accuracy and provides a template for feature extraction from eBPF data.[37]

eBPF at Prompt Security: The First No-Code Security Offering for LLM-Based Apps	Anonymous	2024	Prompt Security Blog	While technical rather than academic, it discusses the use of eBPF to monitor the "entire model stack" in GenAI apps, including API calls and vector database interactions, which parallels monitoring syscalls for container security.[32]

The primary technical constraint identified in these works is that eBPF is not Turing-complete; it prohibits infinite loops and has strict stack size limits.[6] Therefore, the "Student" rules extracted from the Transformer must be highly sparse. This confirms the necessity of using methods like Rule Lists or shallow Decision Trees which can be represented as a series of if-else statements in C.[6]

Area 4: Temporal Logic Mining from Execution Traces

Traditional machine learning models often treat syscall traces as a bag of features or a simple vector, potentially losing the temporal constraints (e.g., "Event A must happen before Event B") that define correct system behavior.[10] Temporal Logic Mining (TLM) addresses this by extracting formal specifications like Linear Temporal Logic (LTL).[38]

Formal Specification Mining Research

Paper Title	Authors	Year	Venue	Summary and Relevance
A Systematic Literature Review on	Anonymous	2025	Software & Systems Modeling	This massive review (2000-2024)

Mining LTL Specifications				covers methodologies for automatically generating LTL from execution traces. It identifies "Anomaly and Malicious Code Detection" as a key use case and discusses automata learning as a core technique.[38]
ARTINALI#: Efficient Intrusion Detection for Cyber-Physical Systems	M. Raiyat Aliabadi et al.	2021	International Journal of Critical Infrastructure Protection	ARTINALI# discovers a small set of security monitors and specifications that yield full attack coverage. It uses a Bayesian network to predict the probability of coverage, reducing memory and runtime overhead by 52% and 69% respectively.[39]
ARTINALI:	Anonymous	2017 (Cited in	ResearchGate	This

Dynamic Invariant Detection for Cyber-Physical System Security		2024)		foundational paper introduces the concept of mining system properties by incorporating time as a first-class property. It significantly reduces false positives (30%) and false negatives (90%) compared to other invariant detection tools.[40]
MAVFI: Micro Aerial Vehicles Resilience Analysis Methodology	Anonymous	2023	DATE Conference	Proposes low-overhead anomaly detection based on Gaussian statistical models and autoencoders. It focuses on the temporal transition of inter-kernel states, which is relevant for modeling the "kinematics" of syscall sequences.[41]

The use of LTL for container security allows for the definition of "safe" execution envelopes. For

example, a mined specification might state: $G(\text{execve} \rightarrow F(\text{close}))$, meaning if a process executes, it must eventually close its file descriptors. Violations of these liveness or safety properties can be detected at runtime by an eBPF monitor.[38] The systematic review emphasizes that while tools like Texada and TACLE are effective, scalability to massive log data remains a challenge, a gap that the Transformer-to-Rule-List architecture could potentially bridge.[38]

Area 5: Bayesian Rule Lists and Uncertainty-Aware Learning

In high-stakes security domains, a model's ability to communicate its confidence is as important as its prediction accuracy. Bayesian Rule Lists (BRL) and probabilistic models provide a framework for uncertainty-aware decision-making.[43]

Bayesian and Probabilistic Rule Learning Research

Paper Title	Authors	Year	Venue	Summary and Relevance
Bayesian Rule Sets for Interpretable Classification	H. Yang et al.	2017 (Cited in 2023/2024)	ResearchGate	This work introduces Bayesian Rule Sets (BRS), a framework for learning short, human-checkable rules. It uses priors that allow users to set a desired model "size and shape," making it highly relevant for kernel deployment where instruction space is

				limited.[45]
A New Prior for Bayesian Anomaly Detection	Anonymous	2022	DBMI Research	Describes a Bayesian prior in the context of pattern detection, combining data learning with semi-informative priors. The algorithm's linear time complexity makes it suitable for real-time monitoring of high-frequency events.[43]
Uncertainty-Aware Bayes' Rule: Concept, Implementation, and Applications	Anonymous	2023	ArXiv	This theoretical work explores a generalized Bayes' rule to combat model uncertainties in prior and data distributions. It suggests applications in classification where balancing the relative importance of belief and data is critical.[46]
Expected Anomaly	Lorenzo Perini et al.	2023	OpenReview / NeurIPS	Proposes a method to rank

Posterior (EAP): An Uncertainty-Based Score Function				the quality of auxiliary anomalies using uncertainty. It addresses the challenge of distinguishing realistic anomalies from noise, which is vital for training syscall IDS on imbalanced datasets.[47]
Learned Manufacturing Inspection Inferences from Image Recognition	Douglas Eddy et al.	2022	FAIM	While applied to manufacturing, this paper argues for Bayesian Rule Lists as a subset of XAI that can learn probabilities of binary occurrences (e.g., attack vs. normal) from sets of observed conditions with high accuracy and interpretability.[48]

The primary innovation of Scalable Bayesian Rule Lists (SBRL) is the use of a generative model that yields a posterior distribution over possible decision lists.[49] Instead of greedy splitting (like traditional CART decision trees), SBRL uses a Bayesian framework to find an optimal list that balances accuracy and sparsity.[44] This is particularly relevant for "Uncertainty Routing," where

sequences with low confidence can be sent to a complex Transformer model in userspace, while certain cases are handled by the lightweight eBPF rules.[46]

Area 6: Distribution Drift and OOD Detection for Security Systems

A deployed IDS must remain resilient to changes in the underlying system behavior (concept drift) and novel attacks that differ from the training data (out-of-distribution).[52]

Monitoring Reliability and Drift Research

Paper Title	Authors	Year	Venue	Summary and Relevance
Concept Drift Detection Based on Decision Distribution in Inconsistent Information System	Anonymous	2023	ResearchGate	Introduces the GraphDrift-net , achieving 99.7% accuracy in detecting drift points in text/log streams. It utilizes the decision distribution function to identify changes in long-term data distributions, relevant for monitoring container workloads.[52]
CAShift: Benchmarking Log-Based Cloud Attack	Anonymous	2025	FSE	This paper introduces a normality shift-aware

Detection under Normality Shift				dataset specifically for cloud attack detection. It shows that most current methods fail to detect container runtime vulnerabilities (like CVE-2024-21626) when the data distribution shifts.[54]
Impact of Concept Drift on Android Malware Detection	Anonymous	2025	ArXiv	Examines virtual and real concept drift, where the characteristics of malware evolve over time. It demonstrates that neural networks offer better resilience to drift due to their latent feature spaces, which supports using a Transformer as the "Teacher".[56]
Generalized Out-of-Distrib	Anonymous	2023	ResearchGate	Provides a unified

ution Detection: A Unified Framework				framework for OOD, anomaly, novelty, and outlier detection. It addresses the gap in recent technical developments and is essential for designing a "fail-safe" mechanism when the rule-based IDS encounters unknown behavior.[53]
Intrusion Detection Model of UAV System Based on Machine Learning	Anonymous	2026	Frontiers in Physics	While focused on UAVs, it highlights how machine learning models can be adapted to new datasets and attack scenarios through transfer learning and continuous monitoring.[57]

The "Normality Shift" concept described in CAShift is especially pertinent to containerized applications.[54] Since Kubernetes often migrates architectures or updates application versions, the "normal" syscall sequence for a specific microservice may change. A feedback loop, as suggested in DeepLog [14], allows for incremental updates to the model weights or rule sets based on live user feedback, ensuring that the IDS adapts without requiring a full redeployment.[14]

Synthesis of XAI Surveys, Baselines, and Literature Gaps

To establish the state-of-the-art for a master's thesis, one must contextualize the work within existing surveys and established performance baselines.

Survey Papers on XAI and Cybersecurity (2022+)

Survey Title	Venue	Year	Key Findings	Source
Explainable AI in Cybersecurity: A Survey	IEEE Access	2022	Reviews 300+ papers; identifies XAI as a "double-edged sword" that improves trust but can be exploited for adversarial attacks.	[19]
Systematic Review of XAI-Enhanced IDS	Frontiers in AI	2025	Notes critical gaps in standardizatio n and scalability; emphasizes the need for hybrid models and real-time explainability.	[29]
Adversarial XAI (Adv-XIDS) Analysis	Info. MDPI	2026	Investigates 135 studies; explores how XAI impacts both deep learning and shallow	[18]

			architectures for Industry 5.0.	
Explainable AI for Internet Traffic Classification and IDS	IEEE Surveys	2024	Comprehensive review of XAI in network traffic analysis; highlights the surge in demand for transparent AI.	[58]

Most-Cited Baselines for Syscall Anomaly Detection

A robust evaluation must compare the proposed Transformer-distilled rules against these established methodologies:

1. **DeepLog (LSTM):** The primary benchmark for log sequence modeling using natural language processing techniques.[14]
2. **STIDE (Sequence Time Delay Embedding):** A classic, simple observable method based on short sequences of system calls.[59]
3. **Isolation Forest:** The leading unsupervised baseline for real-world deployment due to its efficiency and lack of labeling requirements.[61]
4. **ADFA-LD / UNM Public Datasets:** These are the universal standards for testing syscall IDS, despite their age.[60]
5. **Random Forest (Supervised):** Often used as the performance ceiling (e.g., 94.3% accuracy) that explainable models aim to match.[62]

Literature Gaps and Opportunities for Contribution

Based on the review of over 100 research snippets, several critical gaps have been identified that this master's thesis is uniquely positioned to fill:

- **Fidelity-to-eBPF Constraint Optimization:** While Knowledge Distillation into Decision Trees exists [31], there is a lack of research on optimizing the distillation process specifically for the **eBPF verifier's constraints** (e.g., minimizing branching factor and stack depth during the student's training phase).
- **Uncertainty-Aware Kernel Hooks:** Most eBPF enforcement tools make a hard allow/deny decision. No existing work demonstrates an eBPF hook that can calculate the **Bayesian posterior probability** (as in BRL) in-kernel to trigger "selective prediction" or partial enforcement.
- **Multi-modal Kernel Context:** Most syscall models treat the call as a single token. There is

a gap in using **eBPF "X-ray vision"** [5] to feed rich context (process parent, file descriptor owner) directly into a Transformer's multi-modal embedding layer at runtime.

- **Normality Shift in Kubernetes Context:** While CAShift [54] provides a dataset, there is limited work on **feedback loops** that can update eBPF-level rule lists dynamically as a container environment migrates or scales.
- **Neuro-Symbolic eBPF Rules:** Integrating MITRE ATT&CK patterns into the distillation process so that the resulting eBPF rules are not just statistically valid but **symbolically meaningful** to a security analyst.

Conclusion

The intersection of Transformer architectures and eBPF-based enforcement represents a significant shift toward proactive, interpretable, and high-performance container security. The literature confirms that while Transformers excel at capturing the complex, long-range dependencies of system call sequences, their "black-box" nature remains the primary hurdle for operational deployment. Knowledge Distillation to surrogate models like Decision Trees and Bayesian Rule Lists offers a mathematically grounded approach to bridge this gap, allowing for the generation of "white-box" rules that fit within the strict architectural constraints of the Linux kernel. By incorporating temporal logic mining and uncertainty-aware learning, the next generation of intrusion detection systems can provide not only high-accuracy detection but also forensic clarity and adaptive resilience in the face of evolving cloud-native threats. This research area remains fertile for innovation, particularly in the optimization of extracted rules for real-time kernel execution and the automation of security policies in dynamic orchestration environments.

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